

ANOVA

Chapter 13 from Walpole book

Assumptions and Hypotheses in One-Way ANOVA

It is assumed that the k populations are independent and normally distributed with means $\mu_1, \mu_2, \dots, \mu_k$ and common variance σ^2 . As indicated in Section 13.2, these assumptions are made more palatable by randomization. We wish to derive appropriate methods for testing the hypothesis

$$H_0: \mu_1 = \mu_2 = \dots = \mu_k,$$

H_1 : At least two of the means are not equal.

Let y_{ij} denote the j th observation from the i th treatment and arrange the data as in Table 13.2. Here, $Y_{i.}$ is the total of all observations in the sample from the i th treatment, $\bar{y}_{i.}$ is the mean of all observations in the sample from the i th treatment, $Y_{..}$ is the total of all nk observations, and $\bar{y}_{..}$ is the mean of all nk observations.

Table 13.2: k Random Samples

Treatment:	1	2	...	i	...	k	
	y_{11}	y_{21}	...	y_{i1}	...	y_{k1}	
	y_{12}	y_{22}	...	y_{i2}	...	y_{k2}	
	$\vdots$	$\vdots$		$\vdots$		$\vdots$	
	y_{1n}	y_{2n}	...	y_{in}	...	y_{kn}	
Total	$Y_{1.}$	$Y_{2.}$	...	$Y_{i.}$	...	$Y_{k.}$	$Y_{..}$
Mean	$\bar{y}_{1.}$	$\bar{y}_{2.}$	...	$\bar{y}_{i.}$	...	$\bar{y}_{k.}$	$\bar{y}_{..}$

Theorem 13.1: Sum-of-Squares Identity

$$\sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{..})^2 = n \sum_{i=1}^k (\bar{y}_{i.} - \bar{y}_{..})^2 + \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2$$

Three Important
Measures of
Variability

$$SST = \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{..})^2 = \text{total sum of squares,}$$

$$SSA = n \sum_{i=1}^k (\bar{y}_{i.} - \bar{y}_{..})^2 = \text{treatment sum of squares,}$$

$$SSE = \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2 = \text{error sum of squares.}$$

Neil. Weis book formula

Sum of squares	Defining formula	Computing formula
Total, SST	$\Sigma(x_i - \bar{x})^2$	$\Sigma x_i^2 - (\Sigma x_i)^2/n$
Treatment, SSTR	$\Sigma n_j (\bar{x}_j - \bar{x})^2$	$\Sigma (T_j^2/n_j) - (\Sigma x_i)^2/n$
Error, SSE	$\Sigma (n_j - 1) s_j^2$	$SST - SSTR$

Where

n = total no. of observations

n_j = size of j th sample

T_j = sum of j th sample

Sum of Squares,
Unequal Sample
Sizes

$$SST = \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{..})^2, \quad SSA = \sum_{i=1}^k n_i (\bar{y}_{i.} - \bar{y}_{..})^2, \quad SSE = SST - SSA$$

One-Way ANOVA Identity: The total sum of squares(TSS) equals the treatment sum of squares(SSA) plus the error sum of squares(SSE): $TSS = SSA + SSE$.

- SST measures the total variation among all the sample data.
- s_1^2 measures the variation among the samples means.
- s^2 measures the variation within the samples.

The **F-statistic** compares the variation among the sample means to the variation within the samples.

One-Way ANOVA Test

Purpose To perform a hypothesis test to compare k population means, $\mu_1, \mu_2, \dots, \mu_k$

Assumptions

1. Simple random samples
2. Independent samples
3. Normal populations
4. Equal population standard deviations

Step 1 The null and alternative hypotheses are, respectively,

$$H_0: \mu_1 = \mu_2 = \dots = \mu_k$$

H_a : Not all the means are equal.

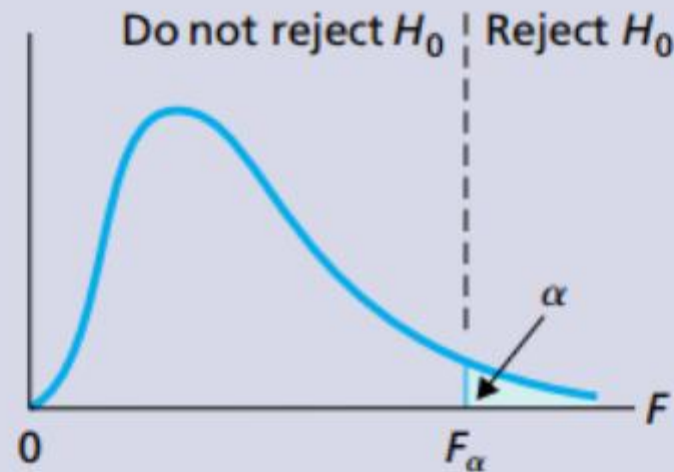
Step 2 Decide on the significance level, α .

Step 3 Compute the value of the test statistic

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	Computed f
Treatments	SSA	$k - 1$	$s_1^2 = \frac{SSA}{k - 1}$	$\frac{s_1^2}{s^2}$
Error	SSE	$k(n - 1)$	$s^2 = \frac{SSE}{k(n - 1)}$	
Total	SST	$kn - 1$		

CRITICAL-VALUE APPROACH

Step 4 The critical value is F_α with $df = (k - 1, n - k)$. Use Table VIII to find the critical value.



Step 5 If the value of the test statistic falls in the rejection region, reject H_0 ; otherwise, do not reject H_0 .

Step 6 Interpret the results of the hypothesis test.

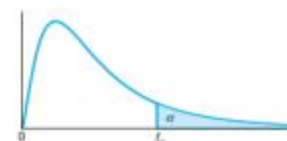


Table A.6 Critical Values of the F-Distribution

$f_{0.05}(v_1, v_2)$									
v_2	v_1								
	1	2	3	4	5	6	7	8	9
1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88

Table A.6 (continued) Critical Values of the F -Distribution

v_2	$f_{0.05}(v_1, v_2)$									
	v_1									
	10	12	15	20	24	30	40	60	120	∞
1	241.88	243.91	245.95	248.01	249.05	250.10	251.14	252.20	253.25	254.31
2	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.36
6	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
∞	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

In Exercises 16.42–16.47, we provide data from independent simple random samples from several populations. In each case,

- compute SST, SSTR, and SSE by using the computing formulas given in Formula 16.1 on page 726.
- compare your results in part (a) for SSTR and SSE with those in Exercises 16.24–16.29, where you employed the defining formulas.
- construct a one-way ANOVA table.
- decide, at the 5% significance level, whether the data provide sufficient evidence to conclude that the means of the populations from which the samples were drawn are not all the same.

16.42

Sample 1	Sample 2	Sample 3
1	10	4
9	4	16
	8	10
	6	
	2	

16.43

Sample 1	Sample 2	Sample 3
8	2	4
4	1	3
6	3	6
		3

16.44

Sample 1	Sample 2	Sample 3	Sample 4
6	9	4	8
3	5	4	4
3	7	2	6
	8	2	
	6	3	

16.45

Sample 1	Sample 2	Sample 3	Sample 4	Sample 5
7	5	6	3	7
4	9	7	7	9
5	4	5	7	11
4		4	4	
		8	4	

16.46

Sample 1	Sample 2	Sample 3	Sample 4	Sample 5
4	8	9	4	3
2	5	6	0	6
3	5	9	2	9

16.47

Sample 1	Sample 2	Sample 3	Sample 4
11	9	16	5
6	2	10	1
7	4	10	3

16.47 (solution on excel)

16.47

Sample 1	Sample 2	Sample 3	Sample 4
11	9	16	5
6	2	10	1
7	4	10	3

decide, at the 5% significance level, whether the data provide sufficient evidence to conclude that the means of the populations from which the samples were drawn are not all the same.

- 1) H_0 : all four population means are equal
- 2) H_1 : atleast two population means are not equal
- 3) $\alpha = 0.05$
- 4) Critical value and critical region

$$F_{\text{critical}} = F_{0.05, v_1, v_2} = F_{0.05, 3, 8} = 4.07$$

critical region is $F > 4.07$

- 5) Test Statistic: (F – test)

Working below

	sample1	sample2	sample 3	sample4	total					
	11	9	16	5	41				grand mean	7
	6	2	10	1	19					
	7	4	10	3	24					
Total	24	15	36	9	84					
mean	8	5	12	3	28					
	SST							798-84^2/12		210
	121	81	256	25						
	36	4	100	1						
	49	16	100	9						
total	206	101	456	35	798					
								SST	210	
	SSA				total					
yi.^2	576	225	1296	81	2178					
	SSA	2178/3-84^2/12	138							
	SSE	210-138	72							

Formulae used:

$$SST = \sum x_i^2 - (\sum x_i)^2 / n$$

$$SSA = \sum (T_j^2 / n_j) - (\sum x_i)^2 / n$$

Where

n = total no. of observations

n_j = size of j th sample

T_j = sum of j th sample

$$SSE = SST - SSA$$

ANOVA Table

		sum of squares	d.f	mean ss	F-ratio
	SSA	138	3	46	5.11111111
	SSE	72	8	9	
Total	SST	210	11		

test statistic value		5.1111			critical value	4.07
		6) conclusion	reject H_0			